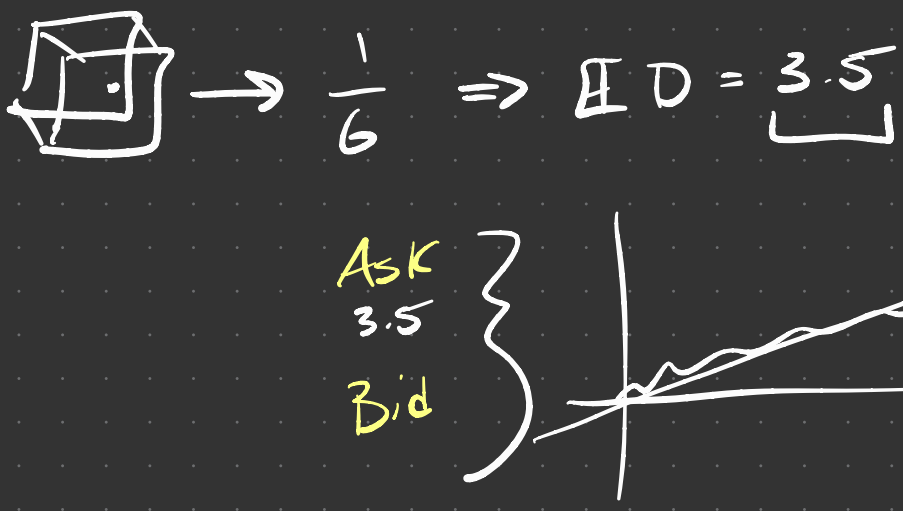



## Defn: Random Experiment

A procedure or process whos outcome can't be predicted

- Coin
- Dice roll
- Roulette

## Defn: Probability Model

A probability model is a math description of a Random experiment that specifies outcomes and assigns probabilities to different "events"

- What outcomes are possible?
- How likely is each event?

- How does knowing one event impact the prob. of another
- What is the exp. value of a random qty?

## Defn: Sample Space

The set of all possible outcomes of a Random experiment

e.x.) Sample Space Dice Roll

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

$$\Omega^2 = \Omega \otimes \Omega = \{(1,1), (1,2), \ldots$$
$$(2,1), (2,2), \ldots$$
$$\vdots$$
$$(6,1), (6,2), \ldots\}$$

## Defn: Event

An event is a subset of the sample space.

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

"Even Dice Rolls" $= E = \{2, 4, 6\}$

$$E \subseteq \Omega$$

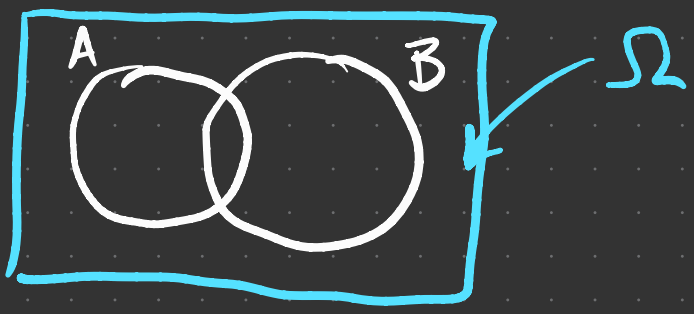
## Defn: Operations on Sets

Given two events $A, B \subseteq \Omega$

Union: $A \cup B = U$

Intersection: $A \cap B = I$

Complement: $A^c = \Omega \setminus A$




## Defn: Probability Measure

A probability measure, $P$, is a function that assign a number to each event and it satisfies the following axioms:

Axiom 1: Nonnegativity

$$P(A) \geq 0$$

Axiom 2: Normalization

$$P(\Omega) = 1$$

Axiom 3: Countable Additivity




pairwise mutually exclusive events

$\forall i, j \quad A_i \cap A_j = \emptyset$

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

$$0 \leq P(A) \leq 1 \quad , \quad \forall A$$

## Rule 1: Complement Rule

$$P(A^c) = 1 - P(A)$$

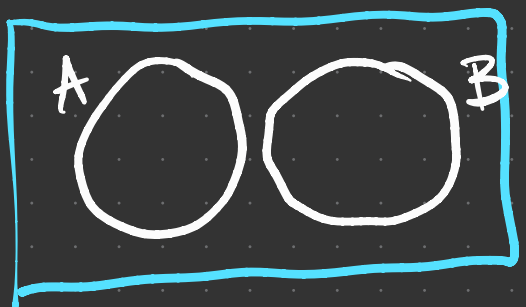
$$P(A^c) + P(A) = 1$$

$$A^c \cup A = \Omega$$

## Rule 2: Addition Rule

For any two events $A, B$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$




$P(A \cap B) = 0$

$A \cap B = \emptyset$

# Defn) Equally Likely Outcomes

A sample space has equally likely outcomes if every event has the same probability.

Ex) $N$ outcomes, each has prob. $\frac{1}{N}$

$$P(A) = \frac{\text{Number of favorable outcomes}}{\text{The total \# of outcomes}}$$

E - "Even dice roll" $= \{2, 4, 6\}$

$$P(E) = \frac{|E|}{|\Omega|} = \frac{3}{6} = \frac{1}{2}$$

Ex) Roll two dice independently.

S - "At least one six"; $P(S) = \frac{|S|}{|\Omega|} = \frac{11}{36}$

$|\Omega| = 36$, $|S| = 11$

$$\frac{25}{36} = P(S^c) \Rightarrow P(S) = 1 - P(S^c)$$

$$= 1 - \frac{25}{36} = \frac{11}{36}$$

# Combinations + Permutations

## Permutation Formula

$$P(n,r) = \frac{n!}{(n-r)!} \qquad \begin{matrix} n=6 \\ r=3 \end{matrix}$$

$$\frac{6!}{(6-3)!} = \frac{6!}{3!} = \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1}$$

## Combinations Formula

$${}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

A B C D E F

6

$${}_6C_3 = \binom{6}{3} = \frac{6!}{3!(3)!} = \frac{120}{3!}$$

$$= \frac{120}{6} = 20$$

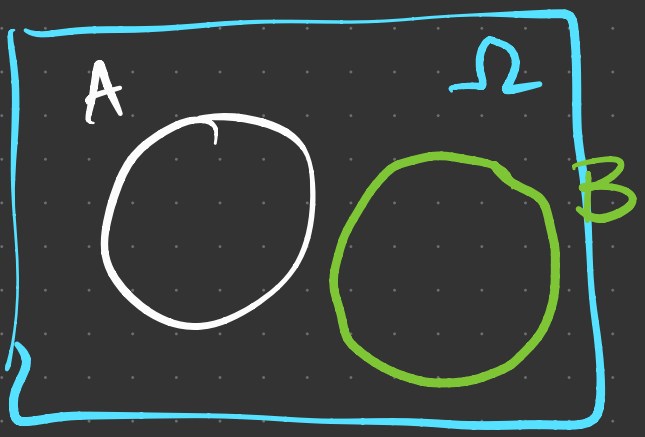
## Defn: Conditional Probability

Given events $A, B$

The cond. prob $A$ given $B$ occured

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$P(B) > 0$




## Multiplication Rule

$$P(A \cap B) = P(A|B)P(B)$$

## Defn: Partition

A collection of events $B_1, B_2, \ldots$ forms a partition if

1) $\forall i,j \; B_i \cap B_j = \emptyset$

2) $\bigcup_{i=1}^{\infty} B_i = \Omega$

## Law of Total Probability

If $B_1, B_2, \ldots$ form a partition

$$P(A) = \sum_{i=1}^{\infty} P(A \mid B_i) P(B_i)$$

## Bayes' Theorem

For a partition $B_1, B_2, \ldots$

$$P(B_i \mid A) = \frac{P(A \mid B_i) P(B_i)}{\sum_{j=1}^{\infty} P(A \mid B_j) P(B_j)}$$

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A \mid B) P(B)}{P(A)}$$

## Defn: Independent Events

Given $A$ and $B$, the events are independent ($A \perp B$) if

$$P(A \cap B) = P(A) P(B)$$

Given $A \perp B$, $P(A|B) = ?$

$$P(A|B) = \frac{P(A)P(B)}{P(B)}$$

$$\Rightarrow P(A|B) = P(A) \text{ if } A \perp B$$

## Defn Mutual Exclusivity

Given events $A$ and $B$,
the events are mutually exclusive
if $A \cap B = \emptyset$ or $P(A \cap B) = 0$

ex.) Given $A \cap B = \emptyset$

$$\Rightarrow P(A \cap B) = 0$$

$$\Rightarrow P(A)P(B) > 0$$

If $A, B$ ME $\Rightarrow$ Not Independent

$P \Rightarrow Q$

## Defn Mutual Independence

A collection of events $A_1, A_2, \ldots, A_n$
is mutually independent if for
every non-negative subset

$$P(A_{i_1} \cdots A_{i_k}) = \prod_{j=1}^{k} P(A_{i_j})$$

## Defn: Random Variable

A random variable (RV) is a real-valued function defined on the outcomes of a random experiment

$$X : \Omega \to \mathbb{R} \quad \text{or} \quad X : S \to \mathbb{R}$$

sample space

$$\Omega = \{HH, TH, HT, TT\}$$

| Outcome ($\omega$) | $X$ |
|---|---|
| HH | 2 |
| TH | 1 |
| HT | 1 |
| TT | 0 |

$$P(X=0) = \frac{1}{4}, \quad P(X=2) = \frac{1}{4}$$

$$P(X=1) = \frac{2}{4} = \frac{1}{2}$$

## Defn: Discrete Random Variable

A discrete RV takes values in a finite or countably infinite set.

## Defn: Prob. Mass Function (PMF)

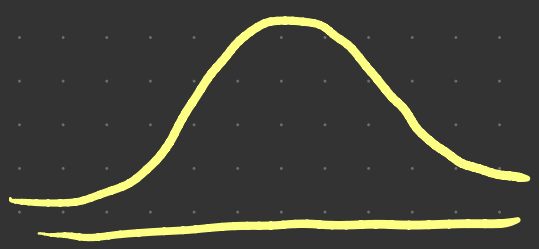
$$P_X(x) = P(X = x)$$

$$P_X(x) \geq 0$$

$$\sum_{a \in A} P_X(a) = 1$$

## Defn: Continuous Random Variable

A continuous RV has a probability distribution that can be described by a probability density function (PDF)



## Defn: Prob. Density Function

A function $f_X(x)$ is a PDF if

$$f_X(x) \geq 0$$

$$\int_{-\infty}^{\infty} f_X(x)\,dx = 1$$

$$\Rightarrow P(a \leq X \leq b) = \int_a^b f_X(x)\,dx$$

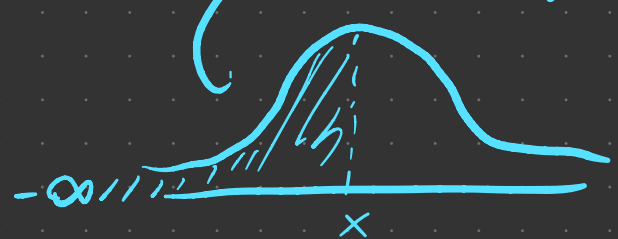
## Defn: Cumulative Distribution Function

The CDF of a random variable $X$ is

$$F_X(x) = P(X \leq x)$$

### Cont. CDF

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(t)\,dt$$



# Essential Probability Distributions

| Distribution | PMF / PDF | Mean | Variance |
|---|---|---|---|
| $X \sim Ber(p)$ | $P(X=1) = p$ | $p$ | $p(1-p)$ |
| $X \sim Bin(n,p)$ | $\binom{n}{k} p^k (1-p)^{n-k}$ | $np$ | $np(1-p)$ |
| $X \sim Geo(p)$ | $p(1-p)^{k-1}, k \geq 1$ | $\frac{1}{p}$ | $\frac{(1-p)}{p^2}$ |
| $X \sim Pois(\lambda)$ | $\frac{e^{-\lambda}\lambda^k}{k!}$ | $\lambda$ | $\lambda$ |
| $X \sim U(a,b)$ | $\frac{1}{b-a}, (a,b)$ | $\frac{a+b}{2}$ | $\frac{(b-a)^2}{12}$ |
| $X \sim Exp(\lambda)$ | $\lambda e^{-\lambda x}, x \geq 0$ | $\frac{1}{\lambda}$ | $\frac{1}{\lambda^2}$ |
| $X \sim N(\mu, \sigma^2)$ | $\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ | $\mu$ | $\sigma^2$ |

# Defn: Expected Value

The expected value of a RV $X$ is its probability weighted sum of outcomes.

Discrete RV: $\mathbb{E}[X] = \sum_x x \, p_X(x)$

Contin. RV: $\mathbb{E}[X] = \int_{-\infty}^{\infty} t \, f_X(t) \, dt$

# Defn: Variance

$$\text{Var}(X) = \mathbb{E}\left[(X-N)^2\right]$$

$$= \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

# Defn: SD

$$\sigma_X = \sqrt{\text{Var}(X)}$$

# Properties of $\mathbb{E}$, Var

RV $X$; $a, b \in \mathbb{R}$

$$\mathbb{E}[aX+b] = \mathbb{E}[aX] + \mathbb{E}[b]$$
$$= a\mathbb{E}[X] + b$$

$$\mathrm{Var}(aX+b) = \mathrm{Var}(aX) + \mathrm{Var}(b)$$
$$= \mathrm{Var}(aX) + 0$$
$$= a^2 \mathrm{Var}(X)$$

Given $X, Y$.

$$\mathbb{E}[X+Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

by linearity of $\mathbb{E}$, regardless if $X \perp Y$

# Defn: Covariance

Given RVs $X, Y$

The covariance measures linear co-variation

$$Cov(X,Y) = \mathbb{E}\left[(X-\mathbb{E}[X])(Y-\mathbb{E}[Y])\right]$$

$$= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Given $X, Y$ if we observe the var.

$$Var(X+Y) = Var(X) + Var(Y) + 2Cov(X,Y)$$

# Defn: Joint Prob. Distribution

Describe the probabilities associated with two or more RVs simultaneously

For discrete $X, Y$ the joint PMF

$$P_{X,Y}(x,y) = P(X=x, Y=y)$$

## Defn: Marginal Distribution

$$P_X(x) = \sum_y P_{X,Y}(x,y)$$

$$P_Y(y) = \sum_x P_{X,Y}(x,y)$$

## Defn: Independent Random Variables

Given $X, Y$ if the joint distribution is a product of the marginals, $X \perp Y$

$$P_{X,Y}(x,y) = P_X(x) P_Y(y)$$

$$f_{X,Y}(x,y) = f_X(x) f_Y(y)$$

## Defn: LLN

Let $X_1, X_2, \ldots \overset{iid}{\sim}$.

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$$

The weak LLN says

$$\bar{X}_n \xrightarrow{P} \mu \quad \text{as} \quad n \to \infty$$

## Defn: CLT

Let $X_1, \ldots, X_n$ i.i.d random variables with $\mu, \sigma^2$

$$\frac{\sum_{i=1}^{n} X_i - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} N(0,1)$$

as $n \to \infty$

$$\bar{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$